# COLLEGE HELPDESK CHATBOT
## Improved Rule-Based Chatbot in Python

### 1. Introduction
This project is a college helpdesk chatbot made with Python conditional rules. It responds to common questions about college services and uses keyword matching with a small fuzzy-match feature for some spelling variations.

### 2. Objective
To practise Python functions, strings, if-elif-else logic, loops, and simple chatbot conversation handling.

### 3. Technologies
Python 3, if/elif/else, string processing, regular expressions, and Python's built-in difflib module. No external Python packages are required.

### 4. Supported Topics
Greetings and help; admissions; courses and syllabus; fees; exams, hall tickets and results; attendance; library; scholarships; hostel; transport; class timetable; faculty; certificates; placements and internships; Wi-Fi and labs; contacts; holidays; student documents; and complaints.

### 5. Working
The chatbot normalizes the message, checks it against predefined keyword rules, selects the closest supported intent, and prints the matching answer. If no rule matches, it asks the user to rephrase or type help. The chat ends when the user types bye, exit, or quit.

### 6. Sample Conversations
The following are example interactions to demonstrate the intended behaviour. Actual answers depend on the keywords entered.

### 7. Limitations
A rule-based chatbot cannot answer every possible question. It does not use a language model, access a college database, or fetch live schedules, fee balances, contact details, and notices. The fallback response is expected for questions outside its rules.

## 8. How to Run

Download college_chatbot.py, open Command Prompt or a terminal in its folder, and run: python college_chatbot.py. Type help to see topics and bye to exit.

## 9. Conclusion

The project demonstrates a practical beginner-level chatbot using Python rules. Its coverage can be extended by adding verified FAQs or connecting it to an approved data source.

## Sample Chatbot Conversations

**You: Hi**

Bot: Hello! Welcome to the College Helpdesk. You can ask me about admissions, courses, fees, exams, attendance, library, scholarships, timetable, departments, certificates, hostel, transport, or contacts.

**You: How do I apply for admission?**

Bot: For admission, check your college's official admission notice. Commonly required documents include previous marksheets, school leaving/transfer certificate, ID proof, photos, and category/income certificates if applicable. Requirements and deadlines vary, so confirm with the admission office.

**You: How can I check attendance?**

Bot: Check your latest attendance in the student portal or with your faculty advisor. Attendance rules vary by institution and subject. If you share classes attended and total classes, I can calculate your percentage.

**You: Tell me about scholarship**

Bot: Scholarship eligibility depends on scheme, income, category, course, and marks. Check the relevant official government/college scholarship portal and confirm deadlines with the scholarship cell.

**You: Tell me a joke**

Bot: I couldn't confidently match that question to my rules. Try rephrasing it or type 'help'. I can answer common college-helpdesk questions, but I don't have live college records or real-time notices.

**You: Bye**

Bot: Goodbye! Best wishes for your studies.

**Submission Checklist**

- Run and test the Python file.
- Fill in student details in the report.
- Upload the Python file, index.html, and README.md to GitHub.
- Publish index.html using GitHub Pages or another static hosting service for a public live link.
- Add the real GitHub and live links before uploading the PDF to GMIU.